

SOKODE, Togo

WMO: 653610

Lat: 8.983N

Lon: 1.150E

Elev: 387

StdP: 96.76

Time Zone: 0.00 (E00)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.7	18.8	-2.1	3.3	26.4	-0.1	3.9	27.1	5.4	25.2	5.0	26.0	0.6	280	0.246

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.9	36.9	22.4	36.0	22.2	35.2	22.1	26.8	32.7	26.3	31.9	25.9	31.3	1.9	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.3	21.4	29.6	24.9	20.9	29.2	24.5	20.4	28.5	86.9	32.7	84.4	32.0	82.5	31.5	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
5.5	4.5	3.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			9.7	28.2	0.8	1.1	9.1	29.0	8.6	29.6	8.2	30.2	7.7	31.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	27.0	28.7	29.3	28.5	27.6	26.5	25.4	24.9	25.4	26.4	26.9	26.3
	DBStd	1.84	1.50	1.35	1.28	1.51	1.40	1.26	1.19	1.02	1.09	1.07	1.05	1.35
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6162	526	523	598	557	544	494	476	461	462	509	507	504
	CDD18.3	3120	267	290	340	307	286	244	218	202	212	251	257	246
	CDH23.3	30363	2934	3602	4272	3556	2962	2044	1354	1093	1313	2081	2612	2540
	CDH26.7	13693	1462	1955	2297	1717	1276	708	332	211	349	839	1317	1233
Wind	WSAvg	1.1	1.2	1.3	1.3	1.2	1.0	0.9	0.8	1.0	0.9	0.9	1.0	1.3
Precipitation	PrecAvg	1357	8	17	56	105	132	174	223	264	242	110	19	4
	PrecMax	2067	220	190	216	181	283	400	420	496	398	329	163	69
	PrecMin	851	0	0	0	42	56	74	18	60	91	20	0	0
	PrecStd	258	33	33	42	35	47	64	78	93	77	63	35	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.7	37.8	38.1	37.0	35.2	33.6	32.0	30.7	32.0	34.1	35.7	35.8
		MCWB	20.4	21.8	22.5	25.2	25.7	26.2	25.0	24.7	25.3	23.8	21.3	20.2
	2%	DB	35.2	36.7	36.9	35.7	34.0	32.2	30.7	29.8	30.8	33.0	34.8	34.7
		MCWB	19.6	21.4	22.8	25.0	25.5	25.4	24.7	24.3	24.9	24.4	21.8	20.0
	5%	DB	34.2	35.8	36.0	34.6	33.0	31.2	29.5	28.8	29.8	32.1	34.0	33.7
		MCWB	19.1	21.3	22.6	25.1	25.5	25.1	24.4	24.0	24.7	24.6	22.0	19.8
	10%	DB	33.1	34.8	35.0	33.5	32.1	30.3	28.5	27.9	28.8	31.1	33.0	32.6
		MCWB	18.6	21.0	22.7	25.2	25.3	24.9	24.1	23.8	24.4	24.5	22.4	19.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.5	26.5	27.0	27.4	27.5	27.0	26.1	25.5	26.2	26.8	25.7	24.9
		MCDB	32.9	34.6	34.9	33.8	32.7	32.3	30.2	29.1	30.4	31.8	31.8	32.0
	2%	WB	23.3	25.2	26.1	26.7	26.7	26.2	25.2	24.7	25.5	25.8	24.9	23.9
		MCDB	31.1	32.7	33.4	32.9	31.9	30.7	29.1	28.2	29.5	30.6	31.1	30.7
	5%	WB	22.1	24.4	25.4	26.1	26.1	25.6	24.7	24.4	25.0	25.3	24.4	22.9
		MCDB	29.5	31.7	32.4	32.3	31.3	29.9	28.3	27.6	28.8	30.1	30.6	29.5
	10%	WB	20.8	23.6	24.8	25.6	25.6	25.1	24.3	24.0	24.6	24.9	23.9	21.8
		MCDB	29.1	30.7	31.5	31.6	30.6	29.3	27.7	27.0	28.0	29.5	30.0	28.3
Mean Daily Temperature Range	5% DB	MDBR	13.7	13.0	11.9	10.1	9.4	8.3	7.1	6.6	7.7	10.1	13.7	14.2
		MCDBR	15.2	14.4	13.5	11.8	10.6	9.7	8.5	8.0	9.1	11.6	14.8	15.5
		MCWBR	6.4	5.5	4.6	3.6	3.5	3.4	3.0	3.0	3.7	4.2	4.5	5.7
	5% WB	MCDBR	14.4	12.7	11.5	10.6	9.9	9.0	7.8	7.3	8.5	10.3	12.6	13.7
		MCWBR	6.9	4.8	4.0	3.9	3.9	4.0	3.8	3.2	3.2	3.8	4.3	4.5
Clear-Sky Solar Irradiance	taub	0.723	0.762	0.815	0.685	0.584	0.535	0.496	0.485	0.492	0.569	0.586	0.621	
	taud	1.589	1.543	1.467	1.727	1.911	2.053	2.207	2.240	2.200	1.945	1.853	1.769	
	Ebn at Noon	633	628	605	682	738	765	801	823	826	760	732	697	
	Edh at Noon	269	290	316	241	196	167	144	142	149	191	204	221	
All-Sky Solar Radiation	RadAvg	5.24	5.41	5.65	5.68	5.49	4.97	4.36	4.08	4.58	5.34	5.50	5.19	
	RadStd	0.17	0.16	0.23	0.16	0.18	0.17	0.17	0.19	0.20	0.20	0.13	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page